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Review

Monitoring External Workload With Wearable Technology After Anterior Cruciate Ligament Reconstruction

A Scoping Review

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Background: Current sports medicine and rehabilitation trends indicate an increasing use of wearable technology. The ability of these devices to collect, transmit, and process physiological, biomechanical, bioenergy, and environmental data may aid in anterior cruciate ligament reconstruction (ACLR) workload monitoring and return-to-sport decision-making. In addition, their ease of use allows assessments to occur outside the clinical or laboratory settings and across a broader timeline.

Purpose: To (1) determine how wearable technology can assess external workload deficits between limbs (involved and uninvolved) and between groups (healthy controls vs patients with ACLR) during physical activity (PA) or sport and (2) describe the types of sensors, sensor specifications, assessment protocols, outcomes of interest, and participant characteristics from the included studies.

Study Design: Scoping review; Level of evidence, 4.

Methods: In February 2023, a systematic search was performed in the MEDLINE, EMBASE, CINAHL, SPORTDiscus, Scopus, IEEE Xplore, Compendex, and ProQuest Dissertations and Theses Global databases. Eligible studies included assessments of PA or sports workloads via wearable technology after ACLR.

Results: Twenty articles met eligibility criteria and were included. The primary activity assessed was activities of daily living, although rehabilitation, training, and competition were also represented. Accelerometers, global positioning system units, pedometers, and pressure sensor insoles were worn to collect external workload data, which was quantified as kinetic, kinematic, and temporospatial data. Daily steps (count) and moderate to vigorous PA (min/day or week) were the most common units of measurement. A limited number of studies included outcomes related to between-limb asymmetries.

Conclusion: The findings of this scoping review highlight the versatility of wearable technologies to collect patients' kinetic, kinematic, and temporospatial data and assess external workload outcomes after ACLR. In addition, some wearable technologies identified deficits in workload compared with healthy controls and between reconstructed and unaffected limbs.

Keywords: wearable technology; anterior cruciate ligament; workload monitoring

Wearable technology describes a diverse group of electronic devices attached to the body or worn by users.[22] Early concepts centered on packaging traditional desktop computing power into smaller information and communication devices (eg, smartphones, smartwatches, and smart eyewear). As the wearable technology market has matured, application to sports and fitness has become one of its most predominant applications.[12,22] Sports and fitness wearable technologies have been embedded into garments and accessories such as watches, bracelets, belts, straps, vests, shirts, pants, and even insoles.[12,22] Smartphones and cloud-based applications instantly allow these devices to collect, transmit, and process physiological, biomechanical, bioenergy, and environmental data.[16]

Workload is commonly used to describe the cumulative quality, quantity, and intensity of physical activity (PA).[26] Workload can be referred to as internal and/or external depending on where the measurable aspects occur.[26] The external workload is measured by the stress applied to the organism (eg, the load lifted or the distance and velocity of running), while the internal workload is measured by the physiological response to the external load (eg, heart rate or neuromuscular or hormonal responses).[26] For the general population, wearable technology provides users with feedback regarding their health and PA, which may help build motivation to achieve a healthier lifestyle.[48] In sports,

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wearable technology has the potential to increase the precision of workload monitoring and fitness tests, which can be helpful when adjusting training loads, avoiding excessive fatigue, and reducing injuries.[16] Wearable technologies may also help monitor patients' workloads after significant injuries. Anterior cruciate ligament (ACL) injuries are particularly fitting because they often require surgery and lengthy rehabilitation.[4] In addition, ACL reconstruction (ACLR) can have poor long-term outcomes, such as a high risk of reinjury, decreased participation in PA or sport, and degenerative changes, including osteoarthritis.[4,13,53]

Previous reviews of wearable technology in sports medicine and rehabilitation indicate that this topic is relatively novel, with research exploring many potential applications.[‖] A recent review of wearable technologies used to identify between-limb deficits during functional tasks after ACLR indicated that wearable technology has the potential to identify patients' kinetic and kinematic deficits.[37] The review included movement assessments of balance, gait via walking or running, change of direction, jumping, and landing.[37] Further development and validation of these wearable technology–enhanced assessments are critical for clinicians facing barriers using the current return-to-sport testing recommendations, especially clinical or laboratory-based assessments.[6,37] Deficits in patients' workload outside a clinical or laboratory setting are also largely unknown and potentially valuable to ACLR return to sport or PA decision-making.[6] Similar to the previous review, this scoping review will focus on external workload measures.

The objectives of this study were to (1) determine how wearable technology can assess external workload deficits between limbs (involved and uninvolved) and between groups (healthy controls vs ACLR) during PA or sport and (2) describe the types of sensors, sensor specifications, assessment protocols, outcomes of interest, and participant characteristics from the included studies.

METHODS

A scoping review was selected because of the broad research question, which required expanded inclusion criteria compared with a systematic review.[39] A scoping review is advantageous because it explores, summarizes, and disseminates research findings and identifies existing evidence gaps.[5] This review follows the 5-stage methodological framework of Arksey and O'Malley[5] and guidance from the Joanna Briggs *Institute Reviewers' Manual*.[44] The PRISMA-ScR (Preferred Reporting Items for Systematic Reviews and Meta-Analyses Extension for Scoping Reviews)[50] guidelines were followed when conducting and reporting this review.

Eligibility Criteria

Return to sport or PA testing is typically conducted between 5 and 10 months after ACLR.[52] However, functional deficits after knee surgeries can persist for up to 2 years in athletes who return to their prior level of sport and longer for athletes who do not return to sport.[36,40] Therefore, this scoping review will include external workload monitoring assessments of PA or sports using wearable technology in perpetuity from surgery. The complete inclusion and exclusion criteria are in Supplemental Table S1, available separately.

Identifying and Selecting Studies

In February 2023, we searched the following electronic databases: MEDLINE, EMBASE, CINAHL, SPORTDiscus, Scopus, IEEE Xplore, Compendex, and ProQuest Dissertations and Theses Global. These databases were searched since inception, with no language limitations. The following search terms were expanded on and used: (wearable technology OR wearable sensor) AND (anterior cruciate ligament) and (physical activity OR workload). In addition, a search for wearable technologies capable of kinetic and kinematic assessment was conducted, and device names as well as product band names were added as search terms. The final list of systematic search terms is in Supplemental Table S2, available separately.

Records obtained from each electronic database were exported into the reference management software Covidence (Veritas Health Innovation; www.covidence.org), where duplicates were removed. At this stage, a single rater (E.G.) independently completed the titles and abstract screening. Two raters (E.G. and A.P.) determined the final study selections by independently performing the remaining full-text reviews with data extraction. Any disagreements on study eligibility between the 2 reviewers were resolved through detailed discussions between the 2 reviewers. The 2 reviewers reached a substantial agreement of 91.39% (Cohen $\kappa = 0.66$; 95% CI, 0.39-0.93).

A descriptive analysis of the included studies was conducted, with wearable technology assessments categorized into activities of daily living (ADL), rehabilitation, and

‖References 6, 14, 17, 27, 37, 43, 45, 47, 54.

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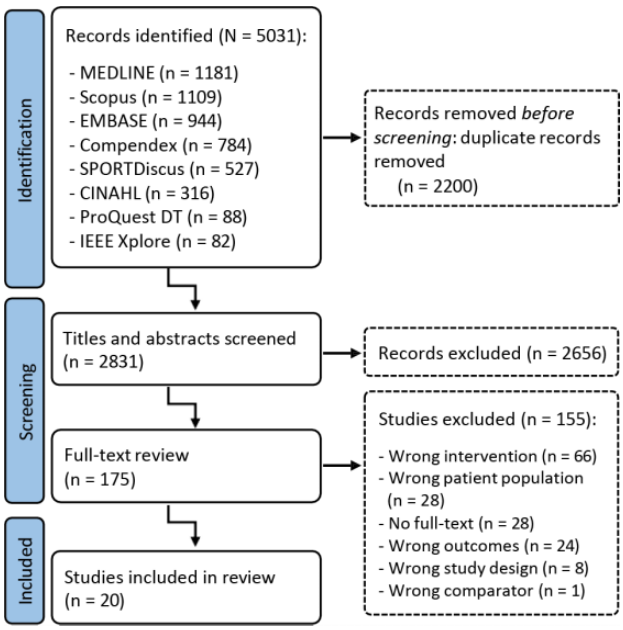

Figure 1. PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) Extension for Scoping Reviews flowchartof included studies.

sports (training or competition). Protocols of wearable technology assessments, outcomes, and participant characteristics were also cataloged within the results. This step allowed descriptive analysis of the utility and appropriateness of different wearable technology devices for a wide range of users during their return to PA or sport after ACLR.

RESULTS

Study Selection

The electronic database search revealed 5031 references for screening, of which 2200 duplicates were removed. After the title and abstract screening of 2831 unique articles, 175 articles proceeded to full-text review. Of these, 20 articles met eligibility criteria and were included in the present scoping review. The systematic search and screening results are presented in a PRISMA-ScR flowchart(Figure 1).

Participants

Among the 20 eligible studies, 935 participants were included, of whom 603 (64%) had undergone ACLR and 332 (36%) were healthy controls. Participants ranged from 12 to 28 years of age (mean, 20.9 ± 4.5 years). The majority of the participants (69%) were female. Six studies recruited only women,[20,21,25,34,49,51] and 1 recruited only men.[35] The majority of participants (77%) were as physically active; 23% were athletes and 1% were elite athletes. The characteristics of the studies are summarized in Table 1.

Postoperative Assessment

There was considerable variation in the postoperative assessment time, ranging from 2 weeks[24,25] to 5 years.[7,8] Four studies conducted assessments in the early to mid-stages of rehabilitation (>6 months postoperatively).[15,24,25,34] Two studies conducted assessments in the later stages of rehabilitation (6-12 months postoperatively).[29,32] One study conducted assessments throughout rehabilitation's early and later stages.[49] Thirteen studies conducted assessments a minimum of 1 year after surgery.[¶]

Wearable Technologies

Accelerometers, global positioning system (GPS) units, pedometers, and pressure sensors were worn to collect external workload (Table 2). The most commonly used product was the GT3X and GT9X Link triaxial accelerometers (ActiGraph).[#] Accelerometers were used across the broadest postoperative time range, from 2 weeks[24,25] to 5.3 years.[8] One study used a simple pedometer, the Digi-Walker SW-200 (New Lifestyles).[7] One study used plantar pressure sensing insoles, the OpenGo (Moticon).[15] Only 1 study utilized GPS technology via the FieldWiz (Insiders).[35]

Assessments

Nineteen studies assessed the external workload of typical daily activities in an uncontrolled free-living environment. These assessment periods were either 1 day,[24,25] 2 days,[15] 7 days,[7,8,10,18,19,28-30,32-34,51] 13 days,[20,21] 2 periods of 28 days,[31] or 231 days.[49] Workloads were also assessed during training activities such as rehabilitation, workouts, and practice[49] and competition, including elite-level rugby matches.[35] Table 3 summarizes the outcomes assessed and the main findings.

Kinematic Outcomes

External workload assessed via daily step count derived from an accelerometer or pedometer's vertical accelerations was the most common kinematic outcome reported in 15 studies.[**] The second most kinematic workload derived from an accelerometer was the duration of moderate to vigorous PA (MVPA), reported as minutes per day or minutes per week, which was reported in 10 studies.[10,18-21,28-30,33,51]

Many of the studies examined between-group external workload differences through kinematic outcomes. Kuenze et al[29] found that adolescents with ACLR (8.0 ± 2.1 months postoperatively) participate in 33% less MVPA (min/day) ($P = .02$) and take 26% fewer daily steps ($P < .001$) than adults with ACLR (8.2 ± 2.1 months postoperatively). As a result, 83% of adults with ACLR and only 9% of adolescents with ACLR achieved age-specific PA guidelines (OR, 60.2; 95% CI, 7.6-493.4).[29]

[¶]References 7, 8, 10, 18–21, 28, 30, 31, 33, 35, 51.
[#]References 8, 10, 18–21, 28–30, 32–34, 51.
[**]References 7, 8, 10, 18, 19, 24, 25, 28, 29, 31–34, 49, 51.

TABLE 1
Study Characteristics (N = 20)[a]

Lead Author (Year)	Participants, n; M/F	Level of PA	Age, y[b]	Height, cm[b]	Weight, kg[b]	Follow-up[b]
Baez (2020)[7]; cross-sectional study	ACLR (RTS) = 15; 4/11 ACLR (NRTS) = 25; 7/18	Physically active	24.3 ± 4.1	169.9 (9.1)	73.2 (15.1)	5 (1-14) y
Barchek (2021)[8]; cross-sectional study	ACLR = 19; 6/13	Physically active	22.9 ± 3.2	170.8 ± 9.1	70.7 ± 10.5	5.3 ± 2.6 y
Bell (2017)[10]; cross-sectional study	ACLR = 33; 11/22 Control = 33; 11/22	Physically active	20.3 ± 1.8 20.8 ± 1.6	171.8 ± 10.5 172.9 ± 8.5	69.9 ± 11.3 70.2 ± 13.5	27.8 ± 17.5 mo
Chan (2021)[15]; cross-sectional study	ACLR =15; 6/9 Control = 15; 6/9	Physically active	26.3 ± 10.8 26.1 ± 10.7	171 ± 8 171 ± 8	71.3 ± 10.3 72.2 ± 1.01	114.8 ± 17.2 d
Davis-Wilson (2022)[18]; cross-sectional study	ACLR (ASY) = 36; 19/17 ACLR (SYM) = 30; 11/19	Physically active	22 ± 4 22 ± 5	171 ± 9 171 ± 9	71.2 ± 14.2 71.5 ± 10.5	39 ± 40 mo 16 ± 15 mo Total: 28 ± 33 mo
Davis-Wilson (2022)[19]; cross-sectional study	ACLR = 31; 14/17 Control = 21; 11/10	Physically active	22 ± 4 23 ± 4	172 ± 9 171 ± 10	71.5 ± 12.8 71.7 ± 14.5	51.9 ± 37.0 mo
Ezzat (2021)[20]; cohort study	ACLR = 51; 0/51 Control = 51; 0/51	Athletes	17.8 (14.9-22.6) 17.8 (14.6-22.1)	166.0 ± 6.4 165.1 ± 5.7	64.0 ± 10.4 61.6 ± 7.7	1.1 (1.0-2.0) y
Ezzat (2022)[21]; cohort study	ACLR = 51; 0/51 Control = 51; 0/51	Athletes	18 (15-22) 18 (15-22)	166 ± 6.4 165.1 ± 5.7	64.0 ± 10.4 61.6 ± 7.7	1.1 (1.0-2.0) y
Gurchiek (2019)[24]; cross-sectional study	ACLR (<6 wk postop) = 6; 3/3 ACLR (>6 wk postop) = 6; 2/4 Control = 16; 8/8	Physically active	26 ± 11 26 ± 6 23 ± 5	174 ± 11 170 ± 13 174 ± 11	70.5 ± 16.2 77.8 ± 15.4 70.5 ± 13.1	ACLR: 2.1 ± 1.6 wk Control: 17.2 ± 2.0 wk
Gurchiek (2019)[25]; cross-sectional study	ACLR (<6 wk postop) = 5; 0/5 ACLR (>14 wk postop) = 5; 0/5	Physically active	26.6 ± 9.1	NR	NR	1.42 (1.1-5.3) wk 15.14 (14.3-19.1) wk
Kuenze (2019[28]; cross-sectional study	ACLR = 31; 8/23 Control = 31; 8/23	Physically active	19.8 ± 1.4 20.6 ± 1.7	172.0 ± 10.6 172.8 ± 9.0	70.2 ± 11.2 69.6 ± 13.7	26.8 ± 15.8 mo
Kuenze (2019)[30]; cross-sectional study	ACLR = 59; 25/34 Control = 55; 22/33	Physically active	20.4 ± 2.3 20.5 ± 1.7	173.0 ± 10.2 174.8 ± 9.4	72.2 ± 12.7 71.5 ± 12.2	29.5 ± 17.9 mo
Kuenze (2021)[31]; feasibility study	ACLR (HSPD) = 6 ACLR (LSPD) = 6	Physically active	22.0 ± 3.0	NR	NR	56.0 ± 36.3 mo
Kuenze (2022)[29]; cross-sectional study	ACLR (ADOL) = 22; 15/7 ACLR (ADLT) = 23; 9/14	Physically active	15.9 ± 1.2 22.5 ± 5.0	170 ± 12 170 ± 13	71.6 ± 19.7 76.0 ± 20.3	8.0 ± 2.1 mo 8.2 ± 2.1 mo
Lisee (2020)[34]; longitudinal study	ACLR = 19; 9/10	Physically active	21.1 ± 5.7	NR	NR	4 mo ± 2 wk
Lisee (2020)[33]; cross-sectional study	ACLR = 57; 23/34 Control = 42; 20/22	Physically active	20.9 ± 3.2 20.7 ± 1.7	180 ± 10 170 ± 10	72.9 ± 12.0 73.3 ± 13.6	28.7 ± 17.7 mo
Lisee (2022)[32]; cross-sectional study	ACLR = 36; 19/17	Physically active	21.4 ± 4.6	NR	NR	8.3 ± 1.7 mo
Lonergan (2018)[35]; cross-sectional study	ACLR = 7; 7/0 Control = 7; 7/0	Elite rugby union	23 ± 3 26 ± 3	186 ± 4 184 ± 7	104 (93-114) 95 (83-104)	458.3 (313-618) d
Taylor (2020)[49]; case report	ACLR = 1; 0/1	Amateur soccer players	12	NR	NR	7-39 wk
Triplett (2021)[51]; cross-sectional study	ACLR = 10; 0/10 Control = 10; 0/10	Physically active	21.4 ± 3.8 21.9 ± 3.1	170.2 ± 5.5 166.0 ± 6.4	75.8 ± 12.8 60.0 ± 7.3	33.0 ± 18.3 mo

[a]ACLR, anterior cruciate ligament reconstruction; ADLT, adult; ADOL, adolescent; ASY, asymptomatic; HSPD, high steps/day; LSPD, low steps/day; M/F, males/females; NR, not reported; NRTS, no return to pre-injury sports participation; postop, postoperative; RTS, return to sport; SYM, symptomatic.
[b]Data are presented as mean ± SD or mean (range).

In 2 studies, Lisee[34] and Lisee et al,[32] authors compared daily step count to laboratory-based kinetic and kinematic gait analysis outcomes (ie, force plates and motion capture). The findings of Lisee[34] suggest that femoral articular cartilage thickness in patients with ACLR at 6 months postoperatively is associated with lesser peak knee extension moment ($\Delta R^2 = 0.20$; $P = .02$) and greater mean steps per day ($\Delta R^2 = 0.19$; $P = .04$) observed 4 months postoperatively. The second study found associations between aberrant gait mechanics and patients with ACLR with low daily step counts.[32] Participants with low daily step counts walked with lesser vertical ground-reaction force during weight acceptance ($d = 1.29$;

TABLE 2
Sensor Specifications[a]

Lead Author (Year)	Wearable Technology	Sampling Rate, Hz	Sensors, n (location)	Analyzed Variables
Baez (2020)[7]	Pedometer (Digi-Walker SW-200; New Lifestyles)	NR	1 (hip)	Daily step (count)
Barchek (2021)[8]	Triaxial accelerometer (GT9X Link; ActiGraph)	NR	1 (wrist)	Daily step (count); mean vertical axis counts per minute (count/min/d)
Bell (2017)[10]	Triaxial accelerometer (GT3X-BT; ActiGraph)	NR	1 (right hip)	Daily steps (count); MVPA (min/d)
Chan (2021)[15]	Plantar pressure sensors (OpenGo; Moticon)	25	13 (insoles)	Total daily limb loading magnitude (N·s/kg); total daily limb loading time (min); between-limb loading asymmetry (%)
Davis-Wilson (2022)[18]	Triaxial accelerometer (GT9X Link; ActiGraph)	NR	1 (right hip)	Daily steps (count); MVPA (min/d)
Davis-Wilson (2022)[19]	Triaxial accelerometer (GT9X Link; ActiGraph)	NR	1 (right hip)	Daily steps (count); MVPA (min/d)
Ezzat (2021)[20]	Triaxial accelerometer (GT3X; ActiGraph)	NR	1 (right hip)	Light PA (min/d); moderate PA (min/d); vigorous PA (min/d); MVPA (min/d)
Ezzat (2022)[21]	Triaxial accelerometer (GT3X; ActiGraph)	NR	1 (NR)	MVPA (min/d)
Gurchiek (2019)[24]	Triaxial accelerometer (BioStamp; MC10) Accelerometer (ActiGraph)	31.25 NR	2 (quadriceps) 1 (waist)	Daily steps (count); walking duration (hours); walking speed (slow/fast); gait asymmetry (%)
Gurchiek (2019)[25]	Triaxial accelerometer (BioStamp; MC10)	31.25	2 (quadriceps)	Daily steps (count); walking duration (hours); gait asymmetry (%)
Kuenze (2019)[28]	Triaxial accelerometer (GT3X-BT; ActiGraph)	NR	1 (right hip)	Daily steps (count); light PA (min/d); moderate PA (min/d); vigorous PA (min/d); MVPA (min/d)
Kuenze (2019)[30]	Triaxial accelerometer (GT3X-BT; ActiGraph)	NR	1 (right hip)	Light PA (min/d); moderate PA (min/d); vigorous PA (min/d); MVPA (min/d)
Kuenze (2021)[31]	Triaxial accelerometer (Charge 3; Fitbit)	NR	1 (wrist)	Daily steps (count); step goal achieved (%)
Kuenze (2022)[29]	Triaxial accelerometer (GT3X-BT and GT9X Link; ActiGraph)	30	1 (right hip)	Daily steps (count); MVPA (min/d, min/wk)
Lisee (2020)[34]	Triaxial accelerometer (GT9X Link; ActiGraph)	30	1 (right hip)	Daily steps (count)
Lisee (2020)[33]	Triaxial accelerometer (GT3X-BT and GT9X Link; ActiGraph)	30	1 (right hip)	Daily steps (count); step cadence (steps/min); MVPA (min/wk)
Lisee (2022)[32]	Triaxial accelerometer (GT9X Link; ActiGraph)	30	1 (right hip)	Daily steps (count)
Lonergan (2018)[35]	GPS (FieldWiz EN, Version A8; Insiders)	10	1 (NR)	Total distance (m); speed (m/min); high-speed meters (5 m/s); accelerations and decelerations (3, 4, and 5 m/s/s)
Taylor (2020)[49]	Accelerometer (ActiGraph) Triaxial accelerometer (VERT; Mayfonk Athletic)	NR NR	1 (waist) 1 (waist)	Daily steps (count); daily distance (m); activity >1 G[b] (min); jumps >15.2 cm (count); maximal jump height (cm)
Triplett (2021)[51]	Triaxial accelerometer (ActiGraph GT9X Link; ActiGraph)	30	1 (right hip)	Daily steps (count); MVPA (min/wk)

[a]CT, contact time; Fmax, maximal ground contact force as percent of body weight; MVPA, moderate to vigorous physical activity; NR, not reported; PA, physical activity.
[b]>1 G = acceleration >1 gravitational force.

95% CI, 0.96-1.62), a stiffened knee strategy, lesser knee extension moment (d = 0.80; 95% CI, 0.48-1.11), and knee flexion angle (d = 1.03; 95% CI, 0.70-1.35) throughout stance compared with those with high daily step counts.[32]

Ezzat et al[20] found no significant differences in MVPA between patients with ACLR (1.1 years postoperatively, approximately 13 months) and healthy controls; however, patients with ACLR had fewer mean minutes per day of vigorous PA compared with the controls (mean difference, –1.22; 95% CI, –2.40 to –0.04]). In a follow-up study, Ezzat et al[21] also found that MVPA was not associated with increased odds of new or reoccurring ACL injury (1.1 years

TABLE 3
Assessments, Protocols, and Main Findings[a]

Lead Author (Year)	Assessments and Protocols	Main Testing Results
Baez (2020)[7]	Pedometer worn during daily activity for 7 d	PA guidelines not met by 29 patients (72%); no significant difference between the RTS and NRTS groups in daily step count; knee self-efficacy and knee-related quality of life were associated with step counts
Barchek (2021)[8]	Accelerometer worn during daily activity for 7 d, ≥10 h/d for at least 4 d (3 weekdays and 1 weekend day)	No significant relationships between fear avoidance beliefs and PA outcomes in patients with ACLR
Bell (2017)[10]	Accelerometer worn during daily activity for 7 d; a minimum of 4 d (3 weekdays and 1 weekend day) of wear for no less than 10 h/d	Patients with ACLR had fewer daily steps and less time in MVPA compared with controls
Chan (2021)[15]	Plantar pressure insoles worn during daily activities for 2 weekdays; ≥9 h/d was required, excluding physical therapy activities	Lower daily limb loading of the surgical vs nonsurgical limb in patients with ACLR
Davis-Wilson (2022)[18]	Accelerometer worn during daily activity for 7 d, ≥10 h/d for at least 4 d (3 weekdays and 1 weekend day)	ACLR (SYM) patients who spent more time in MVPA had a higher quality of life than ACLR (SYM) patients with less time in MVPA
Davis-Wilson (2022)[19]	Accelerometer worn during daily activity for 7 d, ≥10 h/d for at least 4 d (3 weekdays and 1 weekend day)	No significant differences in daily steps or MVPA between patients with ACLRand controls
Ezzat (2021)[20]	Accelerometer worn during daily activity for 13 d, ≥10 h/d for at least 4 d (3 weekdays and 1 weekend day)	No significant difference between groups in MVPA; however, patients with ACLR had fewer mean minutes per day of vigorous PA vs controls
Ezzat (2022)[21]	Accelerometer worn during daily activity for 13 d, ≥10 h/d for at least 4 d (3 weekdays and 1 weekend day)	MVPA was not associated with increased odds of a new or recurrent ACL injury for patients with ACLR or controls
Gurchiek (2019)[24]	Accelerometer worn during daily activity for 1 d	Stride time and composite gait asymmetry decreased across groups: [ACLR <6 wk postop] > [ACLR >6 wk postop] > [controls]
Gurchiek (2019)[25]	Accelerometer worn during daily activity for 1 d	Total walking and fast walking time were significantly greater in ACLR (>14 wk postop) vs ACLR (<6 wk postop) patients; gait asymmetry of the affected and unaffected legs was significantly less in the ACLR (>14 wk postop) vs ACLR (<6 wk postop) patients during slow walking
Kuenze (2019)[28]	Accelerometer worn during daily activity for 7 d, ≥10 h/d for at least 4 d (3 weekdays and 1 weekend day)	No significant relationships between objectively measured PA and patient-reported PA assessments
Kuenze (2019)[30]	Accelerometer worn during daily activity for 7 d, ≥10 h/d for at least 4 d (3 weekdays and 1 weekend day)	The odds of female patients in the ACLR group meeting PA guidelines were worse than that of controls
Kuenze (2021)[31]	Accelerometer worn during daily activity for 28 d (observational), followed immediately by another 28 d under personalized goals setting conditions	Median PA goal achievement was 31.5% ± 6.8% during the intervention; between-group differences in goal achievement were not reported
Kuenze (2022)[29]	Accelerometer worn during daily activity for 7 d, ≥10 h/d for at least 4 d (3 weekdays and 1 weekend day)	At 6-12 mo postop, ACLR (ADOL) patients participated in 33% less MVPA (min/d) and took 26% fewer daily steps vs ACLR (ADLT) patients; as a result, 83% of ACLR (ADLT) and only 9% of ACLR (ADOL) patients achieved age-specific PA guidelines
Lisee (2020)[34]	Accelerometer worn during daily activity for 7 d, ≥10 h/d for at least 4 d (3 weekdays and 1 weekend day)	Smaller knee extension moment (assessed in a laboratory setting) and more daily steps at 4 mo postop were associated with greater medial femoral articular cartilage thickness 6 mo postop
Lisee (2020)[33]	Accelerometer worn during daily activity for 7 d, ≥10 h/d for at least 4 d (3 weekdays and 1 weekend day)	Patients with ACLR had fewer daily steps and walked approximately 40 fewer min/wk in moderate to vigorous intensity cadence compared with controls
Lisee (2022)[32]	Accelerometer worn during daily activity for 7 d, ≥10 h/d for at least 4 d (3 weekdays and 1 weekend day)	Patients with ACLR with the lowest-daily steps walked with an aberrant gait strategy compared with those with higher daily steps
Lonergan (2018)[35]	GPS sensor worn during rugby league match play during either an English Premiership, Championship, or National League One match	The ACLR elite rugby group totaled less distance in a match than controls

(continued)

Table 3 (continued)

Lead Author (Year)	Assessments and Protocols	Main Testing Results
Taylor (2020)[49]	Accelerometer worn during daily activity for 231 d; the second accelerometer was worn during active minutes (MVPA) including rehabilitation, advanced training, and RTS (soccer)	External load metrics at RTS indicated that the most intense rehabilitation session consisted of 36% less frequent movements, 38% lower total distances, and 29% shorter activity durations than the expected demands of a soccer match
Triplett (2021)[51]	Accelerometer worn during daily activity for 7 d, ≥10 h/d for at least 4 d (3 weekdays and 1 weekend day)	Patients with ACLR engaged in fewer daily steps than controls; however, the weekly time spent in MVPA did not differ between groups

[a]ACL, anterior cruciate ligament; ACLR, anterior cruciate ligament reconstruction; ADLT, adult; ADOL, adolescent; ASY, asymptomatic; CoD, change of directions; GPS, global positioning system; MVPA, moderate to vigorous physical activity; NRTS, no return to pre-injury sports participation; PA, physical activity; postop, postoperatively; RTS, return to sport; SYM, symptomatic.

postoperatively, approximately 13 months). Bell et al[10] found that patients with ACLR (27.8 ± 17.5 months postoperatively) had fewer daily steps ($P = .02$; effect size = –0.68) and less time in MVPA than the healthy controls ($P = .02$; $d = –0.72$). Lisee et al[33] found that the patients with ACLR (28.7 ± 17.7 months postoperatively) had fewer daily steps ($P = .005$) and less time in MVPA ($P = .048$) than the controls. Kuenze et al[30] reported that the odds of female patients with ACLR (29.5 ± 17.9 months postoperatively) meeting PA guidelines were significantly worse than that of the healthy controls (OR, 2.54; 95% CI, 1.03 to 6.27). Triplett and Kuenze[51] reported that patients with ACLR (33.0 ± 18.3 months postoperatively) engaged in fewer daily steps than the healthy controls; However, weekly time spent in MVPA did not differ between the groups ($F[1,17] = 6.10$; $P = .02$; $\eta p^2 = 0.26$). In contrast, Davis-Wilson et al[19] found no significant differences in daily steps or MVPA between the patients with ACLR (51.9 ± 37.0 months postoperatively) and healthy controls.

Kinematic Versus Patient-Reported Outcomes

Several studies compared the kinematic outcomes to patient-reported outcomes. Davis-Wilson et al[18] found that patients with osteoarthritis symptoms and ACLR (16 ± 15 months postoperatively) who spent more time in MVPA reported a higher quality of life ($\Delta R^2 = 0.12$; $P = .05$) than nonsymptomatic patients with ACLR. Kuenze et al[28] found no significant relationships between daily steps or PA intensities and self-reported PA or self-reported knee function (26.8 ± 15.8 months postoperatively). Kuenze et al[31] observed daily step count for 28 days and intervened with a personalized step count goal for an additional 28 days (56.0 ± 36.3 months postoperatively). The goal achievement was only 31.5% ± 6.8% during the intervention period.[31] Patient-reported outcomes for self-reported PA or self-reported knee function were not significantly related to changes in daily step count ($P = .63$).[31] Barchek et al[8] found no significant relationships between fear avoidance beliefs and daily steps or vertical counts per minute (5.3 ± 2.6 years postoperatively, approximately 64 months). However, Baez et al[7] observed that patient-reported outcomes related to injury-related fear and self-efficacy accounted for 27.1% of the variance ($P < .004$) of mean daily step counts in patients with a history of ACLR (5 years postoperatively, approximately 59 months).

Temporospatial Outcomes

Gurchiek et al[24] found stride time was significantly greater in the patients with ACLR <6 weeks postoperatively compared with the patients with ACLR >6 weeks postoperatively and healthy controls ($r = –0.91$; $P < .01$). The authors also measured workload with a multioutcome composite asymmetry. They found significantly greater asymmetries in the patients with ACLR <6 weeks compared with the patients with ACLR >6 weeks postoperatively and healthy controls ($r = -0.87$, $P < .01$).[24] Gurchiek et al[25] observed that patients with ACLR >14 weeks walked more often (1.95 ± 0.33 vs 1.34 ± 0.39 hours; $P = .03$) and faster (stride time: 1.14 ± 0.11 vs 1.34 ± 0.10 seconds; $P < .01$) compared with the participants with ACLR<6 weeks. Patients with ACLR <6 weeks postoperatively had significantly greater gait asymmetries during slow walking than patients with ACLR >14 weeks postoperatively ($P < .05$; $d = 9.25$).[25] Taylor et al[49] examined daily steps and estimated daily distance using the participant's mean stride length. In addition, the authors used an alternative method to calculate vigorous activity, which recorded the duration of activities >1 G and jumps >15.2 cm in height.[49] Upon returning to sport (39 weeks postoperatively, approximately 9 months), the participant's external load metrics indicated that the most intense rehabilitation session consisted of 36% less frequent movements, 38% lower total distances, and 29% shorter activity durations than the expected demands of a match.[49] Lonergan et al[35] found that elite rugby players with ACLR (458 days postoperatively, approximately 15 months) totaled less distance in a match than the healthy players ($P = .004$).

Kinetic Outcomes

Only 1 study used wearable technology to measure gait kinetics using plantar pressure sensing insoles.[15] Chan and Sigward[15] examined daily limb loading (N·s/kg) over a 2-day, free-living period (114.8 ± 17.2 days postoperatively,

approximately 3-4 months). The authors reported lower daily limb loading in the surgical limb of patients with ACLR compared with the nonsurgical limb ($P < .001$; $d = 0.63$), as well as the matched limbs of healthy control participants (surgical: $P = .037$, $d = 0.80$; nonsurgical: $P = .02$, $d = 0.89$).[15]

DISCUSSION

The growth of wearable technologies in health care has been slower to advance than in sports and fitness because of barriers such as the need for increased rigor of validity and reliability as well as big data concerns over security and ethics.[54] Still, reviews of wearable technology in health care indicate a growing trend, especially for sports medicine and rehabilitation.[6,37,43,47] This scoping review demonstrates the versatility of wearable technologies in monitoring the external workloads of patients with ACLR. The primary activity assessed in the studies was ADL, although a limited number of activities such as rehabilitation, training, and competitions in sports were also present. The activities assessed via wearable technology in this review would not have been possible in a clinical or laboratory setting because of space and/or time constraints. The paucity of research on external workload during sports training and competition after ACLR is concerning, considering that return to sport is often the primary goal of ACLR rehabilitation.[38]

Clinical or laboratory-based return-to-PA or -sport testing occurs between 5 and 10 months postoperatively, with the most common time of assessment being 6 months, with a return to sport recommended no earlier than 9 months postoperatively.[9,52] This scoping review highlights how wearable technology can greatly expand the assessment time frames and offer insight at virtually any time point throughout the ACLR return-to-PA or -sport process. As a result, practitioners, clinicians, and other stakeholders could access information beyond that obtained in clinical or laboratory-based assessments, which could aid in return-to-PA or -sport decision-making. Previous studies reported that only 23% of patients returning to sport or activities had passed their assessment.[38,52] The decision not to return to sport is complex, including multiple physical and psychological factors; however, half of the athletes who do not return cite lower PA as the primary reason not to return.[38] Patients and clinicians monitoring PA through rehabilitation would help set goals and increase motivation.[48] Additionally, clinicians' between-limb asymmetry outcomes obtained via wearable technology may help direct their rehabilitation practices without needing laboratory-based asymmetry assessments.

Pressure sensor insoles and GPS sensors were underrepresented compared with accelerometers but may present greater opportunities to collect unique data (kinetic between-limb asymmetries) or for use in more dynamic settings (sports). In contrast to this review, GPS has been the most used wearable technology to monitor external workload in sports performance literature.[11] Interestingly, this review found no studies directly stating the use of inertial measurement units (IMUs). IMUs are a popular device in sports science research.[1,2,11] These devices comprise accelerometers, gyroscopes, and magnetometer sensors, allowing for a more comprehensive measure of movement and orientation in space.[3] However, many wearable technology products used in this review were, in fact, IMUs, suggesting that only the accelerometer data were analyzed. It is also possible that the lack of IMU representation is due to a terminological oversight.

This scoping review reveals a wide variety of devices, quantitative outcomes, and the apparent sensitivity to identify deficits in external PA workload between patients with ACLR and healthy controls or between the affected and unaffected limbs. The rapid advancement of wearable technology can offer consumers new opportunities for fitness and health, but further work is needed to determine the validity and reliability of these outcomes in different activity populations.[46] It was beyond the scope of this review to assess the validity and reliability of all outcomes of interest; however, all the represented devices have acceptable criterion validity for measuring step count and PA monitoring.[23,41,42] Criterion validity for wearable technology external workload monitoring in ACLR rehabilitation practices will likely substantially impact their contribution to a successful return to sport or PA. Therefore, at this time, researchers, clinicians, and other stakeholders should exercise caution when interpreting wearable technology outcomes related to ACLR.

Limitations

This scoping review intended to summarize all the existing literature for wearable technology used to assess the external workload of athletes after ACLR (within the confines of the inclusion/exclusion criteria). As the broad range of technologies, assessments, outcomes, and study designs made appropriate weighting of quality assessment questions difficult, a quality assessment was not conducted. In addition, only studies with full-text articles available in the English language were included.

Future Directions

Little detail was provided regarding the activities participants engaged in during ADL assessment periods. Many patients with ACLR will not return to competitive sports, yet may still participate in strenuous, higher-risk activities such as weightlifting, running, hiking, mountain biking, skiing, or snowboarding. Assessments of these activities could provide valuable insight into external workload deficits for a broader range of patients with ACLR. Further investigation into the impact of ACLR on the external workload for competitive athletes in a wider variety of sports would also be helpful. Although not included in this review, internal workload, particularly surface electromyography, may be an area of interest for clinicians. Further validation of wearable technology external workload outcome associations with successful return to sport or PA is needed.

CONCLUSION

This scoping review highlights the versatility of wearable technologies to assess patients' kinetic, kinematic, and temporospatial external workload outcomes after ACLR. These devices appear sensitive enough to identify deficits in external workload compared with healthy controls and between the affected and unaffected limbs of patients with ACLR. In addition, their ease of use allows assessments to occur outside clinical or laboratory settings and over extended timelines. Although between-group (ACLR vs healthy controls) and between-limb (affected vs unaffected) deficits in workload were common, there is no evidence at this time that these assessments could predict future ACL injuries.

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